

GIRRAWEE HIGH SCHOOL

Year 11 Mathematics

Test 3

June 2007

Time: 90 minutes

Instructions:

Start each question on a new page.

Show all necessary working.

Marks will be deducted for careless or badly arranged work.

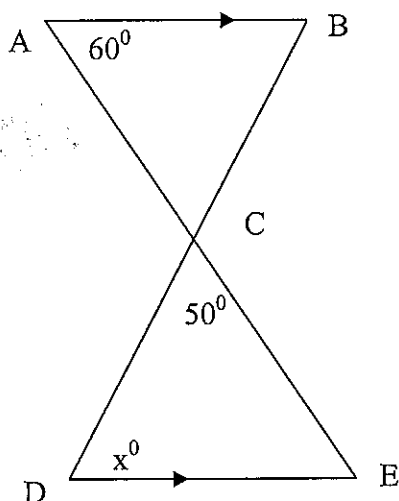
Board approved calculators may be used.

Diagrams are NOT to scale.

Question 1 (10 marks)

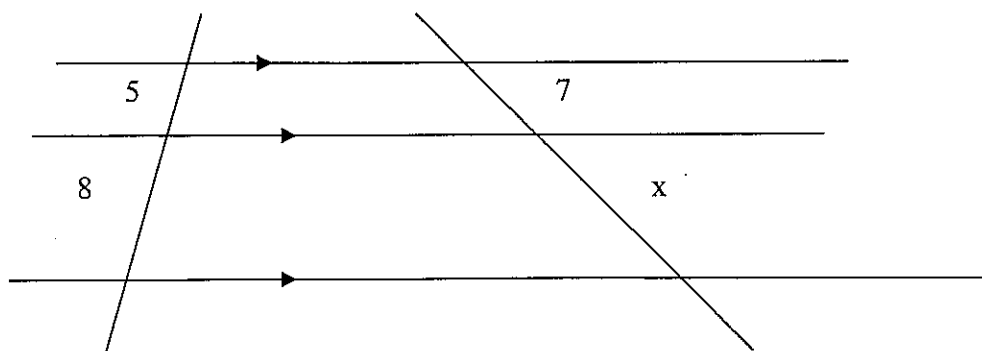
(a) Find x° giving reasons in:

(3 marks)



(b) Find x giving reasons :

(3 marks)



(c) Each exterior angle of a regular polygon is 30° .

(i) Find the number of sides the polygon has.

(2 marks)

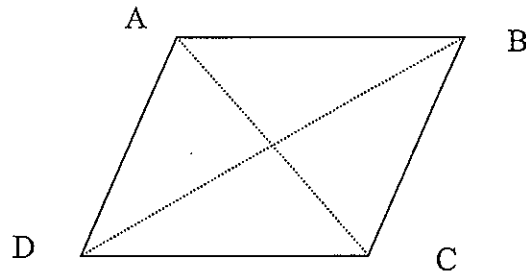
(ii) Find the size of each internal angle of the polygon.

(2 marks)

Question 2 (15 marks)

(a) A triangle has side lengths 7cm, 9cm and 11cm. Prove that it is NOT right-angled. (2 marks)

(b) In rhombus ABCD, $AC = 8\text{cm}$ and $DB = 15\text{cm}$. Find the perimeter of the rhombus. (4 marks)

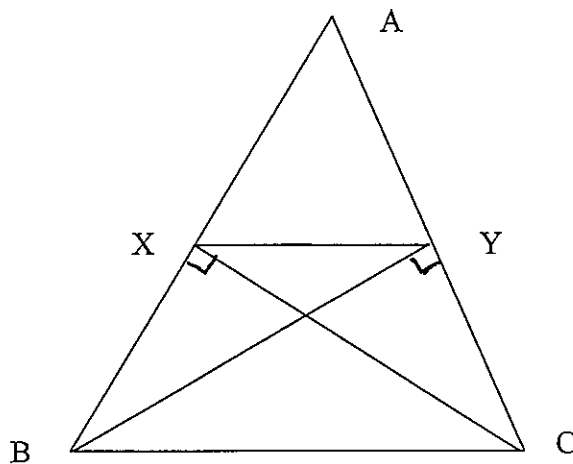


(c) $\triangle ABC$ is an isosceles triangle. $AB = AC$, $BY \perp CA$ and $CX \perp BA$. Prove

(i) $\triangle BXC \equiv \triangle CYB$ (4 marks)

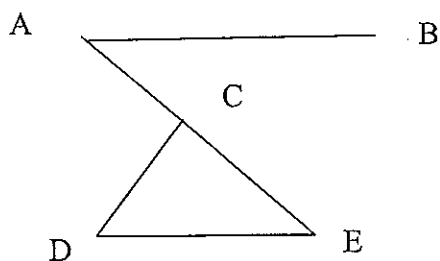
(ii) $AX = AY$ (2 marks)

(iii) $XY \parallel BC$ (3 marks)



Question 3 (15 marks)

In the diagram below, $AB \parallel ED$, $\triangle ECD$ is isosceles with $EC = DC$ and $\angle BAE = 30^\circ$.



(i) Copy the diagram and mark on it all of the given information.

(1 mark)

(ii) Find angle DCE giving reasons..

(3 marks)

(b) For the diagram below:

(i) Prove $\triangle ADB \cong \triangle ABC$.

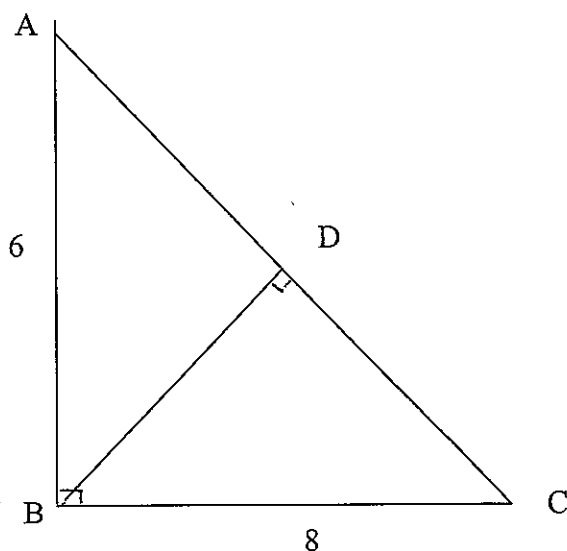
(3 marks)

(ii) Find AC.

(2 marks)

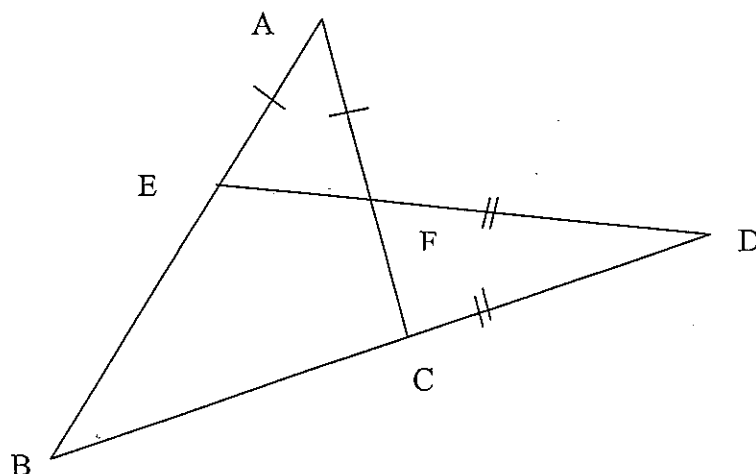
(iii) Find BD.

(3 marks)



(c) If $AE = AF$ and $DC = DF$ in the diagram below, prove $\angle EAF = \angle FDC$

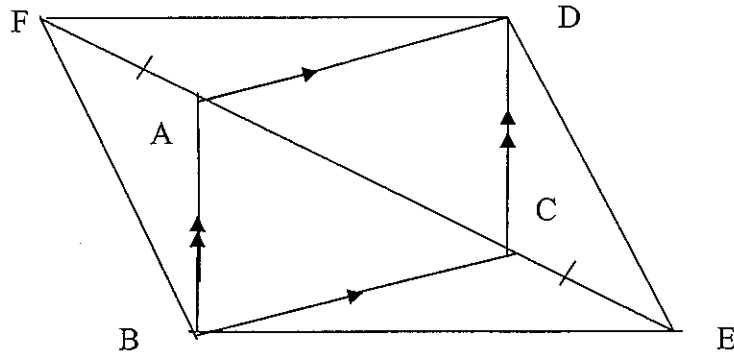
(3 marks)



Question 4

(10 marks)

In the diagram below, ABCD is a parallelogram and $AF = CE$.



- (a) Prove that $\angle FAD = \angle ECB$.
 (b) Prove that $\triangle ADF \equiv \triangle CBE$
 (c) Prove that FBED is a parallelogram.

(3 marks)

(3 marks)

(4 marks)

Question 5 (27 marks)

- (a) For each of the following curves:
 Sketch them on separate number planes
 State their domain and range
 State whether they are functions

(5 marks each)

(i) $y = 3^x$

(ii) $y = |2x - 3|$

(iii) $x^2 + y^2 = 25$

(iv) $y = x^2 + 2$

(v) $y = \frac{1}{x-5}$

- (b) State the DOMAIN of $y = \frac{1}{\sqrt{3x+15}}$

(2 marks)

Question 6 (16 marks)

(a) If $f(x) = x - \frac{1}{x}$ find

(i) $f(2)$

(1 mark)

(ii) $f(a-2)$

(1 mark)

(iii) Find x if $f(x) = 0$.

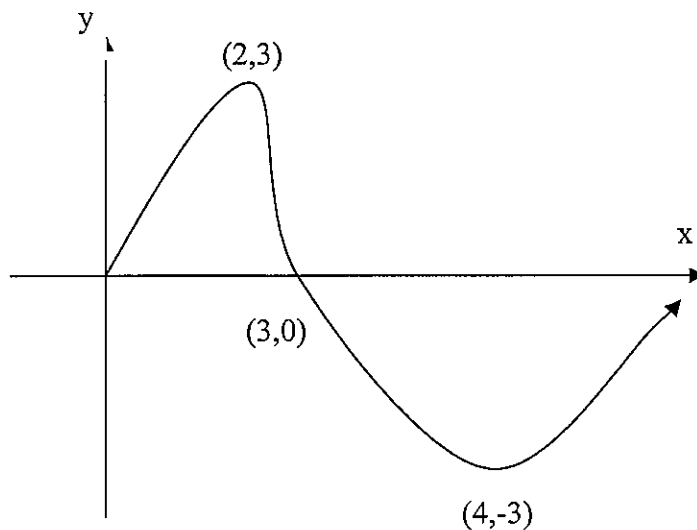
(3 marks)

(b) If $g(x) = \begin{cases} x-2, & x < -1 \\ x^2-4, & -1 \leq x \leq 2 \\ 3x-6, & x > 2 \end{cases}$

Sketch the graph of $g(x)$ for $-2 \leq x \leq 3$ showing all x and y intercepts and points where the graph changes direction. (4 marks)

(c) (i) Prove that the function $f(x) = x^4 - x^2 + 1$ is an even function. (3 marks)

(ii) Copy and complete this graph for $y = h(x)$ for $-5 \leq x \leq 5$ if $h(x)$ is an odd function. (4 marks)



Task 3 2007 Solutions:

Marks:

Q.(1)(a) $\angle AED = 60^\circ$ [alternate $\angle$'s in $\parallel$ lines =].
 $x^\circ = 70^\circ$ [angle sum $\triangle DCE = 180^\circ$].

= (3)

(b) $\frac{x}{7} = \frac{8}{5}$ [$\parallel$ lines cut transversals in same ratio]
 $x = 11\frac{1}{5}$ units

(3)

(c)(i) Number of sides = $360^\circ \div 30^\circ$ [exterior $\angle$'s of polygon add to 360°].
 $= 12$

(2)

(ii) Each internal angle = $180^\circ - 30^\circ$
 $= 150^\circ$ [angles in straight line supplementary]

(2)

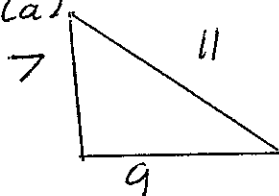
OR total sum of interior angles = 180×10
 $= 1800^\circ$

[interior angles of n sided polygon add to $180(n-2)^\circ$].

Each angle in 12 sided polygon

$= 1800 \div 12$
 $= 150^\circ$

Q.(2)(a)



Triangle is right-angled if,

by Pythagoras' theorem, $c^2 = a^2 + b^2$, c being longest side [hypotenuse if right-angled].

$$11^2 = 7^2 + 9^2$$

$$121 = 130$$

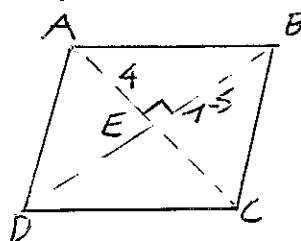
As LHS $\neq$ RHS, triangle is NOT right-angled.

(2)

(b) $AC \perp DB$ [diagonals of rhombus $\perp$]

$AE = EC = 4\text{cm}$ [diagonals of rhombus bisect each other],

$DE = EB = 7.5\text{cm}$ [" "]



Hence $(AB)^2 = 4^2 + 7.5^2$ [by Pythagoras' theorem]
 $= 72.25$

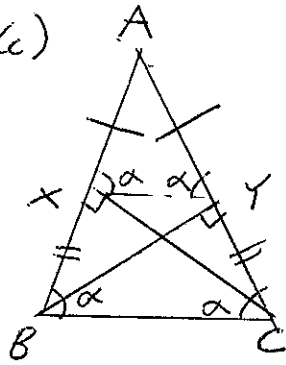
$AB = 8.5\text{cm}$

P of rhombus = 8.5×4

$= 34\text{cm}$

(4)

Q. (2)(c)



(i) Let $\angle ABC = \alpha$.

$$\angle ACB = \angle ABC = \alpha$$

[$\angle$'s opposite = sides in isosceles $\triangle ABC$ =]

BC common

$$\angle BXC = \angle CYB = 90^\circ \text{ [data]}. \quad (4)$$

Hence $\triangle BXC \equiv \triangle CYB$ [AAS].

(ii) $AX = AB - BX$ and $AY = AC - CY$

But $AB = AC$ [data]

and $BX = CY$

[matching sides in $\equiv \triangle BXC$ and $\triangle CYB$]. (2)

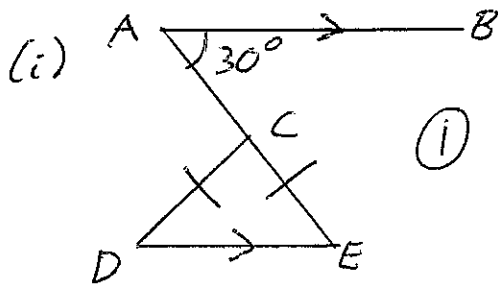
$\therefore AX = AY$.

(iii) $\angle BAC = 180^\circ - 2\alpha$ (angle sum of $\triangle ABC = 180^\circ$).

$$\angle AXY = \angle AYX = \alpha \text{ [}\angle\text{'s opposite = sides in isosceles } \triangle AXY \text{ =]}. \quad (3)$$

Hence $XY \parallel BC$ [angles in corresponding positions =].

Q. (3)(a)

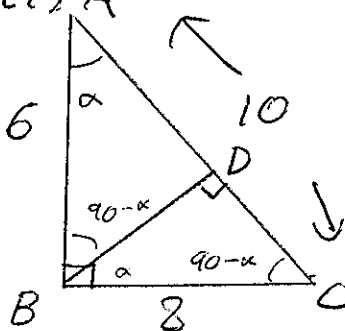


(ii) $\angle DEC = 30^\circ$ [alt. $\angle$'s in $\parallel$ lines AB and DE =].

$$\angle CDE = 30^\circ \text{ [}\angle\text{'s opposite = sides in isosceles } \triangle DEC \text{ =]}.$$

$$\angle DCE = 120^\circ \text{ [angle sum } \triangle DCE = 180^\circ \text{]}. \quad (3)$$

(b)(i)



$\angle A$ common

$$\angle ABC = \angle ADB = 90^\circ \text{ [data]} \quad (3)$$

Hence $\triangle ADB \parallel \triangle ABC$ [all pairs of matching angles equal].

$$(ii) (AC)^2 = 6^2 + 8^2$$

$$AC = 10 \text{ units [By Pythagoras' theorem]}. \quad (2)$$

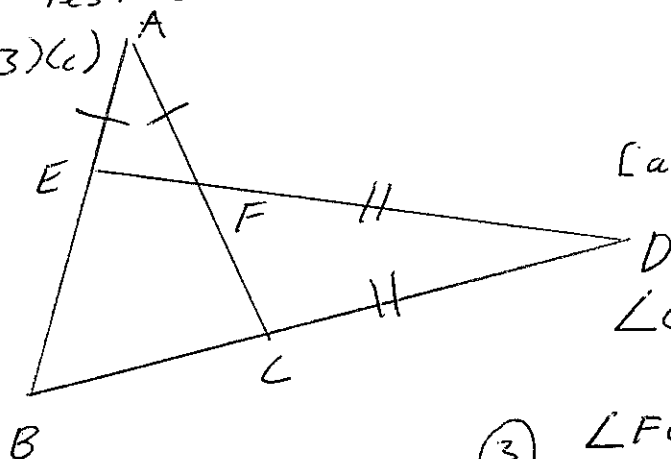
(iii)

$$\frac{BD}{BC} = \frac{AB}{AC} \text{ [matching sides in } \parallel \triangle ADB \text{ and } \triangle ABC \text{]}.$$

$$\frac{BD}{8} = \frac{6}{10}$$

$$BD = 4\frac{4}{5} \text{ units} \quad (3)$$

Q.(3)(c)



Let $\angle EAF = \alpha$

$\angle AEF = \angle AFE = 90^\circ - \frac{1}{2}\alpha$ ✓

[angles opposite = sides in isosceles $\triangle AEF =$].

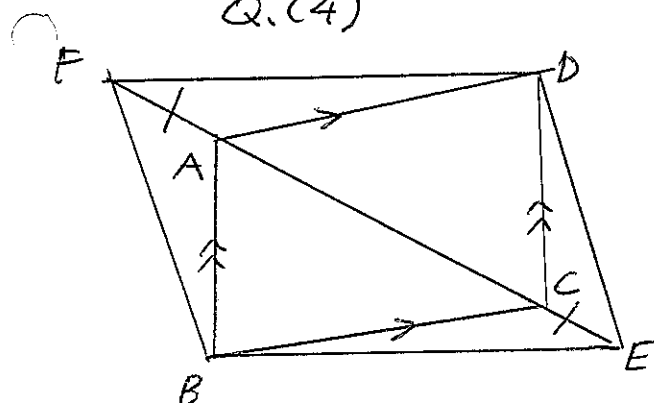
$\angle CFD = 90^\circ - \frac{1}{2}\alpha$ [vertically opposite $\angle$'s =]

(3) $\angle FCD = 90^\circ - \frac{1}{2}\alpha$ [$\angle$'s opposite = sides in isosceles $\triangle FCD =$].

$\angle FDC = \alpha$ [angle sum $\triangle FDC = 180^\circ$]



Q.(4)



(a) Let $\angle FAD = \alpha$

$\angle CAD = 180^\circ - \alpha$ [$\angle$'s in a straight line supplementary].

$\angle ACB = 180^\circ - \alpha$ [alternate $\angle$'s in $\parallel$ lines AD and BC =]

$\angle BCE = \alpha$ [$\angle$'s in a straight line supplementary].

Hence $\angle BCE = \angle FAD$.

(b) $AF = CE$ [data]

$BC = AD$ [opposite sides of parallelogram =].

$\angle FAD = \angle BCE$ [proven in (a)].

$\therefore \triangle ADF \equiv \triangle CBE$ [SAS].

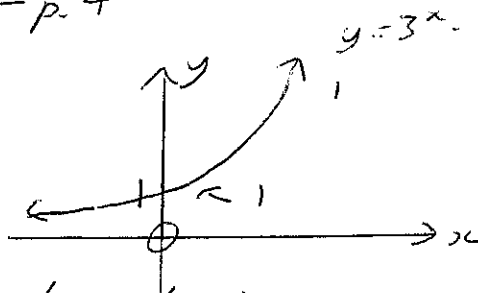
(c) $FD = BE$ [matching sides in $\equiv \triangle ADF$ and $CBE =$].

$\angle DFA = \angle BEC$ [matching $\angle$'s in $\equiv \triangle ADF$ and $CBE =$].

$\therefore FD \parallel BE$ [alternate $\angle$'s =].

Hence FBED is a parallelogram [one pair of opposite sides equal and parallel].

Q. (5)(a)(i)



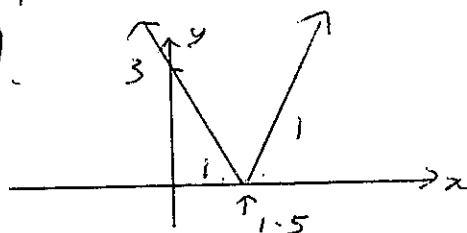
Domain: all real numbers.

Range: $y > 0$.

is a function.

(5)

(ii) $y = |2x - 3|$



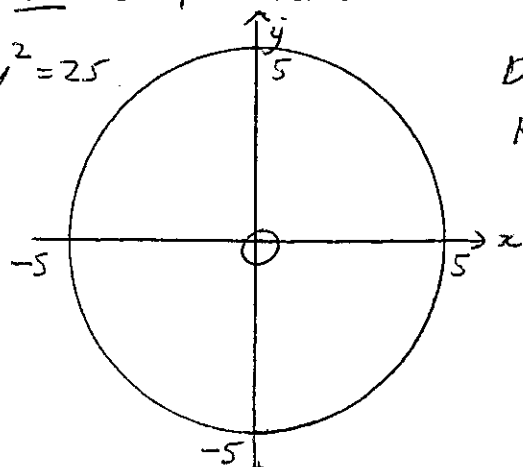
Domain: All real numbers.

Range: $y \geq 0$

is a function.

(5)

(iii) $x^2 + y^2 = 25$



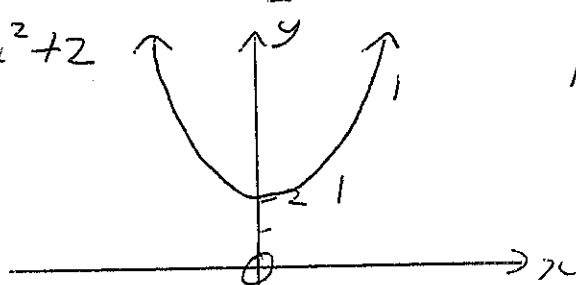
Domain: $-5 \leq x \leq 5$

Range: $-5 \leq y \leq 5$.

is NOT a function.

(5)

(iv) $y = x^2 + 2$



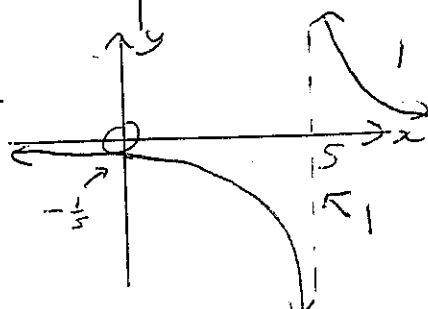
Domain: All real numbers.

Range: $y \geq 2$

is a function.

(5)

(v) $y = \frac{1}{x-5}$



Domain: All real numbers except 5.

Range: All real numbers except 0.

is a function

(5)

(b) Domain of $y = \frac{1}{\sqrt{3x+15}}$ is where $3x+15 > 0$
 $3x > -15$
 $x > -5$.

(2)

Domain: $x > -5$.

Q.16) (a) $f(x) = x - \frac{1}{x}$

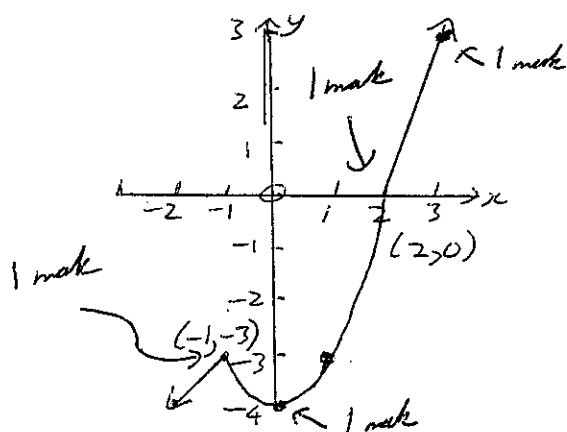
(i) $f(2) = 2 - \frac{1}{2}$
 $= 1\frac{1}{2}$ (1)

(ii) $f(a-2) = a-2 - \frac{1}{a-2}$ (1)

(iii) If $f(x) = 0$
 $x - \frac{1}{x} = 0$
 $x^2 - 1 = 0$
 $(x-1)(x+1) = 0$
 $x = \pm 1$ (3)

(b)

x	-2	-1	0	1	2	3
$g(x)$	$-2-2$ $= -4$	$(-1)^2-4$ $= -3$	0^2-4 $= -4$	1^2-4 $= -3$	2^2-4 $= 0$	3^2-6 $= 3$
		Check: $-1-2$ $= -3$			Check: $3-2-6$ $= 0$	



(c) (i) $f(x) = x^4 - x^2 + 1$ is EVEN $\nabla f(-a) = f(a)$
 LHS: $f(-a) = (-a)^4 - (-a)^2 + 1$
 $= a^4 - a^2 + 1$
 RHS: $f(a) = a^4 - a^2 + 1$
 $= \text{LHS}$ (3)

Hence $f(x) = x^4 - x^2 + 1$ is an even function.

